Parth Hindiya - My Today Work

1 Background Research

- **Read the Official Paper:** Checked out the Monash paper to understand how the `.tsf` file format works and how they use MASE for benchmarking.
- **Figured out the Baselines:** Looked into how the basic models work under the hood:
  - *Naive Model:* Just carries the last known value forward.
  - *Seasonal Naive Model:* Copies data from the same day last week ($m = 7$) to handle the weekly rhythm.

2 Data Loading & Setup

- **Got the Data:** Downloaded the US Births dataset and used the official repo tools to load the `.tsf` file straight into a Pandas DataFrame.
- **Cleaned the Columns:** Picked a simple two-column format with just `date` (as the index) and `birth_rate`.

3 Project Directory

- Set up a prototyping notebook (`.ipynb`) to work with the data and plots.
- Organized the project folders to keep future code separate from the testing code.

4 Exploratory Data Analysis (EDA)

- **Checked Data Quality:** Confirmed the dataset is complete with absolutely zero missing or null values.
- **Spotted Trends:** Noticed a clear pattern showing births drop significantly on weekends, along with regular yearly ups and downs.
- **Found Outliers:** Used the simple IQR formula to flag weird drops (like major holidays):

$$\text{IQR} = Q_3 - Q_1$$

$$\text{Normal Range} = [Q_1 - 1.5 \times \text{IQR},\ Q_3 + 1.5 \times \text{IQR}]$$

5 What's Next

- **Right Now:** Writing the loop for the walk-forward (rolling-origin) backtesting split.
- **Up Next:** Getting the Naive and Seasonal Naive models running so I can fill out the first baseline results table (MAE, RMSE, MAPE).